openbuildingsway

Predict how a building sways in an earthquake, floor by floor.

PAN IIT OSS explainer • diploma / undergraduate to advanced undergraduate (levels 5-6)

Why this exists

A building is a stack of floors held up by springy columns, and when the ground shakes it sways in several patterns at once. This explainer follows the mathematics from one mass on a spring to a ten-storey frame, using only simple matrices and a few lines of code, so you can build it yourself.

The end goal

Predict building sway in an earthquake with multi-DOF vibration. The end goal is predicting building sway in an earthquake with multi-degree-of-freedom vibration: periods, mode shapes, drift and base shear for a shear building, validated against exact results. It is an educational model, never a method for seismic design, which needs qualified engineers, building codes and far richer models.

How the four sessions build up

1. **Session 1: A building is a mass on a spring** (Level 1, conceptual interactive model, 60 min).
2. **Session 2: One storey, one rhythm: resonance and stiffness design** (Level 2, mathematical engineering model, 75 min).
3. **Session 3: Many floors, many modes: a ten-storey frame in an earthquake** (Level 3, simulation and validation, 90 min).
4. **Session 4: The response-spectrum shortcut: base shear without a time history** (Level 3, simulation and validation, 90 min).

1 Session 1: A building is a mass on a spring

A. Real-world problem

After an earthquake some buildings are badly shaken while the one next door hardly moves. Part of the answer is rhythm. Every building has its own slow, regular sway, and the ground has a favourite rhythm too. If the two match, each shove adds to the last, like pushing a child on a swing at just the right moment. What decides a building's rhythm, and how bad is the match? (This repo is for learning. It is a simplified model and must never be used to design or judge a real building.)

B. Concept and theory

Picture the whole building as one lump of mass m on one spring of stiffness k, where k is the sideways force needed per metre of sway. Pull it sideways and let go: it swings back and forth. The angular frequency w = sqrt(k / m) tells how fast the cycle goes in radians per second, and the period T = 2 pi / w is the time for one full back-and-forth sway. Heavier means slower, stiffer means faster. For typical frame buildings engineers often use the rough empirical rule of thumb T

about 0.1 N seconds for N storeys; it is only a guide. When the ground is shaken back and forth with period Tg, the gain says how much the sway is amplified compared with a very slow push. With r = T / Tg and a damping ratio zeta (typically 0.05, meaning the sway dies away by a few percent each cycle), the gain is 1 / sqrt((1 - r^2)^2 + (2 zeta r)^2). Session 2 derives it.

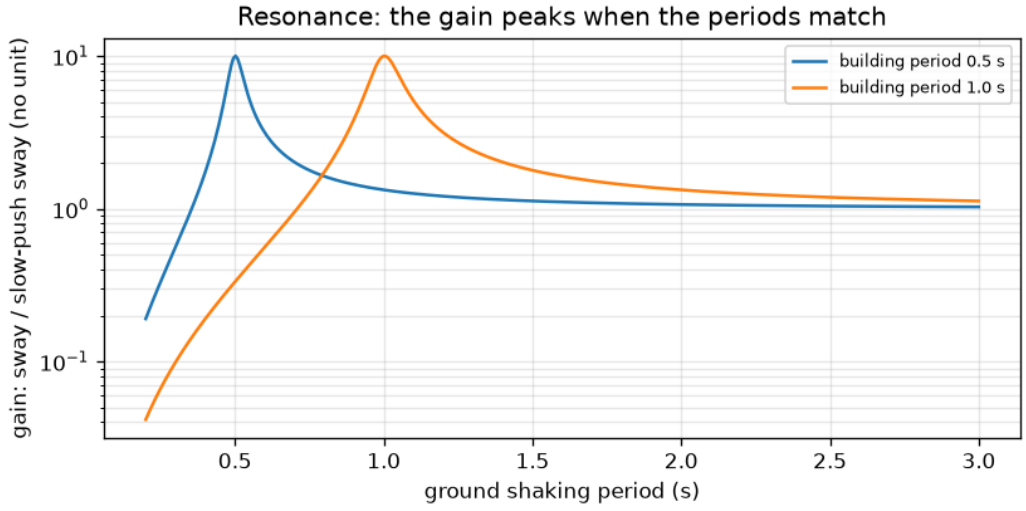
- m: mass of the building lump (kg)
- k: sideways stiffness of the columns (N/m)
- w: angular frequency of the sway (rad/s)
- T: period of the sway (s)
- Tg: period of the ground shaking (s)
- r: ratio T / Tg (no unit)
- zeta: damping ratio (no unit)
- gain: sway divided by the sway for a very slow push (no unit)

$$\omega = \sqrt{\frac{k}{m}}, \qquad T = \frac{2\pi}{\omega}$$

$$x(t) = x_0 \cos(\omega t) \quad \text{(free sway released from rest at } x_0\text{)}$$

$$T \approx 0.1\,N \text{ s} \quad \text{(rough empirical rule of thumb for typical frame buildings)}$$

$$\text{gain} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{T}{T_g}$$



Gain against ground period for buildings of 0.5 s and 1 s: the peak of 10 sits where the periods match.

E. Worked example

A building lump has m = 200 000 kg and k = 20 000 000 N/m. Find w and T. What happens to T if the mass doubles, and if the stiffness is multiplied by four? At resonance with 5 percent damping, how much is the sway amplified?

1. w = sqrt(k / m) = sqrt(20 000 000 / 200 000) = sqrt(100) = 10 rad/s.
2. T = 2 pi / w = 6.283 / 10 = 0.6283 s.
3. Double the mass: T grows by a factor sqrt(2), so T = 0.6283 * 1.4142 = 0.8886 s.

4. Four times the stiffness: T shrinks by a factor sqrt(4) = 2, so T = 0.3142 s.
5. At resonance r = 1, so the gain is 1 / sqrt(0 + (2 * 0.05 * 1)^2) = 1 / 0.1 = 10.

Answer. w = 10 rad/s and T = 0.6283 s; doubling the mass gives 0.8886 s, quadrupling the stiffness gives 0.3142 s; at resonance the gain is 10.

F. Questions

1. **Conceptual.** Why does a heavier building sway more slowly while a stiffer building sways faster? Why can a matching ground rhythm be dangerous even when the push itself is small?
2. **Numerical.** A building lump has m = 500 000 kg and k = 20 000 000 N/m. Find w and T and say whether it is close to the 1 s ground period.
3. **Graph and simulation interpretation.** On the gain graph a 1 s building is shaken by a ground whose period is 2 s, and then by one with period 0.5 s. Read the gain in each case and say what each tells an engineer.
4. **Design and optimisation challenge.** A building has m = 200 000 kg and the ground period is 1 s. Choose a stiffness k that makes its period 0.5 s (a stiff design) or 1.5 s (a flexible design). What does each give for the gain, and what is the downside of each?

2 Session 2: One storey, one rhythm: resonance and stiffness design

A. Real-world problem

Soft soils in many cities shake mostly with a period near 1 s. A one-storey building has a roof mass of 200 tonnes on columns whose stiffness makes its period exactly 1 s. How large does it sway if the ground shakes with an acceleration of 2 m/s^2? And how stiff must the columns be to move the building safely away from that rhythm? (Educational model only, never for real design.)

B. Concept and theory

Let u be the roof displacement relative to the ground. Newton's law for the roof mass m, with a spring k, a damper c and the ground acceleration a_g(t), gives m u" + c u' + k u = -m a_g. Dividing by m and writing w = sqrt(k / m) and zeta = c / (2 m w) gives u" + 2 zeta w u' + w^2 u = -a_g(t). For ground acceleration a0 sin(W t) the sway settles, after a short transient, into a steady wave with amplitude u_max = (a0 / w^2) D(r), where r = W / w is the frequency ratio and D(r) = 1 / sqrt((1 - r^2)^2 + (2 zeta r)^2) is the amplification factor. Since W = 2 pi / Tg and w = 2 pi / T, the ratio is also r = T / Tg. The factor a0 / w^2 is the sway if the acceleration were applied as a constant push. D is largest at r = sqrt(1 - 2 zeta^2) (close to 1) and its maximum is about 1 / (2 zeta); for the typical zeta = 0.05 that is about 10. The rule of thumb T about 0.1 N seconds is a rough empirical guide for N-storey frame buildings.

- m: roof mass (kg)
- k: storey stiffness (N/m)
- c: damping coefficient (N s/m)
- zeta: damping ratio c / (2 m w) (no unit)
- w: natural angular frequency sqrt(k / m) (rad/s)
- T, Tg: natural period of the building, period of the ground shaking (s)
- a0: amplitude of the ground acceleration (m/s^2)
- r: frequency ratio W / w = T / Tg (no unit)
- D: amplification factor (no unit)

- u_max: steady sway amplitude relative to the ground (m)

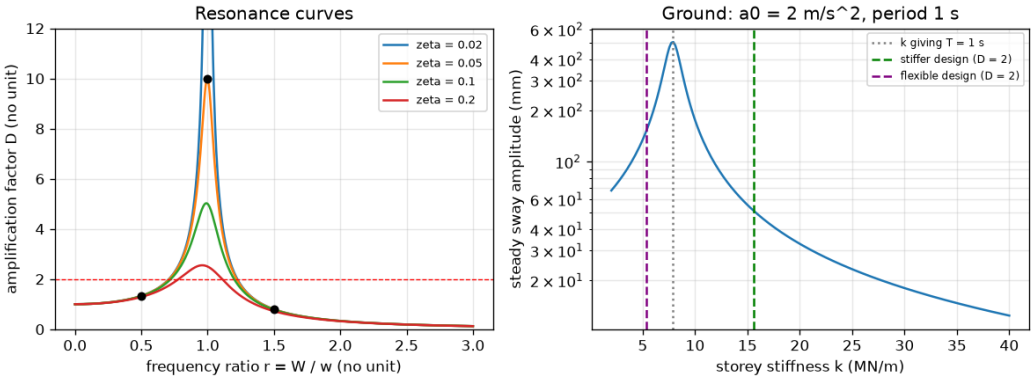
$$m\ddot{u} + c\dot{u} + ku = -m\,a_g(t) \implies \ddot{u} + 2\zeta\omega\dot{u} + \omega^2 u = -a_g(t)$$

$$\omega = \sqrt{k/m}, \quad T = \frac{2\pi}{\omega}, \quad \zeta = \frac{c}{2m\omega}, \quad r = \frac{\Omega}{\omega} = \frac{T}{T_g}$$

$$u_{\max} = \frac{a_0}{\omega^2}\,D(r), \qquad D(r) = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$

$$D_{\max} = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \approx \frac{1}{2\zeta} \quad \text{at } r = \sqrt{1-2\zeta^2}$$

$$k = m\left(\frac{2\pi}{T}\right)^2$$



Left: amplification factor against frequency ratio for four dampings, with numerical dots. Right: steady sway against storey stiffness, with the stiff and flexible design points.

E. Worked example

A roof mass m = 200 000 kg sits on columns of period T = 1 s. (a) Find k. (b) For a ground acceleration amplitude of 2 m/s^2 and a period of 1 s with zeta = 0.05, find the steady sway. (c) How stiff must the columns be so that D is at most 2 on the stiff side?

1. k = m (2 pi / T)^2 = 200 000 * (6.2832)^2 = 200 000 * 39.478 = 7.896e6 N/m, and w = 2 pi / 1 = 6.283 rad/s.
2. At T = Tg = 1 s, r = 1, so D = 1 / (2 zeta) = 10.
3. a0 / w^2 = 2 / 39.478 = 0.05066 m, so u_max = 0.05066 * 10 = 0.5066 m: more than half a metre.
4. For (c) solve D(r) = 2 with r below 1: (1 - r^2)^2 + (0.1 r)^2 = 0.25. This gives r = 0.7107 (r^2 = 0.505, 1 - r^2 = 0.495, and 0.495^2 + 0.0051 = 0.250).
5. The building period must be T = r Tg = 0.7107 s, so k = m (2 pi / 0.7107)^2 = 1.563e7 N/m, about 1.98 times the original stiffness.

Answer. k = 7.896e6 N/m gives resonance with a sway of 0.5066 m for D = 10. Making the columns 1.98 times stiffer (k = 1.563e7 N/m, T = 0.7107 s) brings D down to 2. A much more flexible building also works: D = 2 at r = 1.2186, T = 1.2186 s and k = 5.317e6 N/m (0.673 times the original).

F. Questions

1. **Conceptual.** Why does the amplification factor go above 1 near r = 1 but fall below 1 for large r? What does damping change?
2. **Numerical.** A storey has m = 100 000 kg and k = 4e6 N/m, damping 5 percent. The ground has period 1 s and acceleration amplitude 2 m/s^2. Find w, T, r, D and the steady sway amplitude.
3. **Graph and simulation interpretation.** On the resonance curves, read D at r = 1 for zeta = 0.02, 0.05, 0.1 and 0.2, and D at r = 1.5 for zeta = 0.05. What does each tell you?
4. **Design and optimisation challenge.** The ground period is 1.5 s (softer soil) and the roof mass is 200 000 kg. Find the stiffness and the building period for D = 2 on the stiff side and on the flexible side, with the sway amplitude for a0 = 2 m/s^2. Which design would you prefer and why?

3 Session 3: Many floors, many modes: a ten-storey frame in an earthquake

A. Real-world problem

A real building is not one lump: each floor is a mass, each storey of columns is a spring, and the stack can bend in many patterns at once. For a ten-storey frame with a first period near 1 s, how large is the sway at each floor in a synthetic earthquake record, which storey is bent most, how big is the force at the base, and what happens if one storey is much weaker than the rest? Can we trust the answer? (A simplified, linear, planar model for learning, never for seismic design.)

B. Concept and theory

Number the floors 1 to N from the bottom; floor 0 is the fixed ground. Let u_j be the sway of floor j relative to the ground, m_j its mass, and k_j the stiffness of storey j, between floors j - 1 and j. The equation of motion is M u" + C u' + K u = -M r a_g(t), where M is diagonal, K is tridiagonal (each floor feels only its two neighbours) and r is a vector of ones. Free vibration u = phi cos(w t) gives the generalised eigenproblem K phi = w^2 M phi. Because M is diagonal, put phi = M^(-1/2) v to get the ordinary symmetric problem (M^(-1/2) K M^(-1/2)) v = w^2 v, which np.linalg.eigh solves. The columns phi_i are the mode shapes (scaled so phi^T M phi = 1) and w_i the frequencies. For equal masses m and equal storey stiffness k, with a fixed base and a free top, the standard exact result is w_i = 2 sqrt(k / m) sin((2i - 1) pi / (2(2N + 1))) with shape sin((2i - 1) j pi / (2N + 1)) at floor j; we verify it numerically. The ground motion couples to mode i through the participation factor Gamma_i = phi_i^T M r, and the effective modal mass Gamma_i^2 measures how much of the total mass that mode moves; the effective masses of all modes add up to the total mass. Each mode obeys q" + 2 zeta w q' + w^2 q = -Gamma a_g(t) and u = sum of phi_i q_i (modal superposition). Alternatively the full equation is stepped directly by the Newmark-beta method with the average acceleration parameters beta = 1/4 and gamma = 1/2 (unconditionally stable, no numerical damping). The damping matrix C = M Phi diag(2 zeta w) Phi^T M gives the same zeta in every mode. Design quantities are the inter-storey drift u_j - u_(j-1) (drift ratio = drift / storey height), and the base shear = k_1 u_1, the force in the lowest storey.

- N: number of storeys (no unit)
- m: mass of one floor (kg)
- k: stiffness of one storey (N/m)
- M, K, C: mass, stiffness and damping matrices (kg, N/m, N s/m)

- u: floor sway relative to the ground (vector) (m)
- w_i, phi_i: frequency and mode shape of mode i (rad/s, shape)
- Gamma_i: participation factor of mode i (kg^(1/2))
- a_g(t): ground acceleration (synthetic record) (m/s^2)
- zeta: damping ratio in every mode (0.05) (no unit)
- V: base shear, k1 u1 (N)

$$M\ddot{\mathbf{u}} + C\dot{\mathbf{u}} + K\mathbf{u} = -M\mathbf{r}\,a_g(t)$$

$$K\boldsymbol{\phi} = \omega^2 M\boldsymbol{\phi} \iff (M^{-1/2}KM^{-1/2})\mathbf{v} = \omega^2\mathbf{v}, \quad \boldsymbol{\phi} = M^{-1/2}\mathbf{v}$$

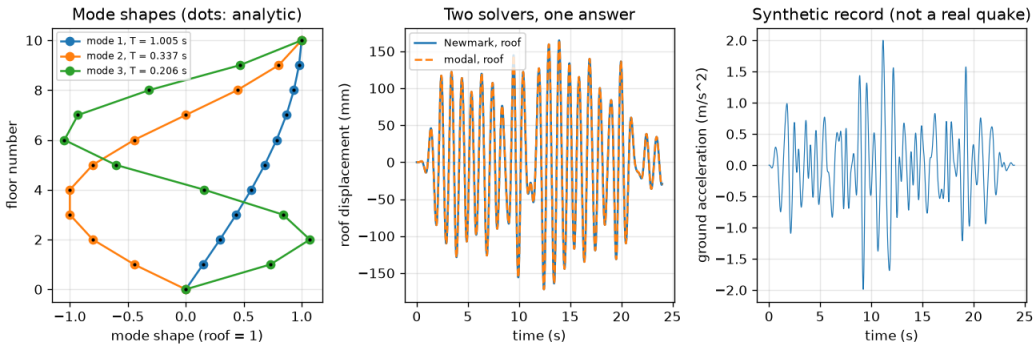
$$\omega_i = 2\sqrt{\frac{k}{m}}\sin\frac{(2i-1)\pi}{2(2N+1)}, \qquad \phi_{j,i} \propto \sin\frac{(2i-1)j\pi}{2N+1}$$

$$\Gamma_i = \boldsymbol{\phi}_i^{\mathsf{T}} M\mathbf{r}, \qquad M_i^* = \Gamma_i^2, \qquad \sum_i M_i^* = \mathbf{r}^{\mathsf{T}} M\mathbf{r}$$

$$\ddot{q}_i + 2\zeta\omega_i\dot{q}_i + \omega_i^2 q_i = -\Gamma_i a_g(t), \qquad \mathbf{u} = \sum_i \boldsymbol{\phi}_i q_i$$

$$\text{Newmark:}\quad \dot{\mathbf{u}}_{n+1} = \dot{\mathbf{u}}_n + \Delta t[(1-\gamma)\ddot{\mathbf{u}}_n + \gamma\ddot{\mathbf{u}}_{n+1}], \quad \beta = \tfrac{1}{4},\ \gamma = \tfrac{1}{2}$$

$$\text{drift}_j = u_j - u_{j-1}, \qquad V = k_1 u_1$$



Left: first three mode shapes (dots: exact formula). Middle: roof sway from Newmark and from modal superposition. Right: the synthetic ground record.

E. Worked example

Find the periods, mode shapes and effective masses of a two-storey building by hand with m = 200 000 kg and k = 3.5e8 N/m, and check them with the code.

1. K = k [[2, -1], [-1, 1]] and M = m I. With lambda = w^2 m / k the eigenproblem is [[2 - lambda, -1], [-1, 1 - lambda]] phi = 0.
2. The determinant condition (2 - lambda)(1 - lambda) - 1 = lambda^2 - 3 lambda + 1 = 0 gives lambda = (3 -/+ sqrt 5) / 2 = 0.382 and 2.618.
3. k / m = 1750 per s^2, so w1^2 = 0.382 * 1750 = 668.4 and w1 = 25.854 rad/s (T1 = 0.2430 s); w2^2 = 2.618 * 1750 = 4581.6 and w2 = 67.687 rad/s (T2 = 0.0928 s).

4. From the first row, phi2 / phi1 = 2 - lambda: 1.618 for mode 1 (both floors move the same way; this is the golden ratio, the same number as in the Fibonacci example of Atlas S22) and -0.618 for mode 2 (the floors move in opposite directions).
5. Participation for equal masses: Gamma^2 / total mass = (phi1 + phi2)^2 / (2 (phi1^2 + phi2^2)). Mode 1: (2.618)^2 / (2 * 3.618) = 0.947. Mode 2: (0.382)^2 / (2 * 1.382) = 0.0528. They add to 1.

Answer. T1 = 0.2430 s with shape (1, 1.618) carrying 94.7 percent of the mass, and T2 = 0.0928 s with shape (1, -0.618) carrying 5.3 percent; the code reproduces all of it. For the 10-storey frame the same code gives T1 = 1.0049 s, T2 = 0.3375 s and T3 = 0.2056 s; the first mode carries 84.8 percent of the mass, the second 9.1 percent and the third 3.1 percent.

F. Questions

1. **Conceptual.** Why does the first mode carry most of the mass in the response, and why does the second mode have a node (a floor that hardly moves)?
2. **Numerical.** Two storeys with m = 100 000 kg and k = 1e8 N/m: find both frequencies and periods by hand, then say which mode is more dangerous for a ground period of 1 s.
3. **Graph and simulation interpretation.** In the soft-storey bar chart, which storey carries the largest drift and how much more than in the uniform frame? What does the damage tell an engineer?
4. **Design and optimisation challenge.** Design: the first storey is weaker than the others by a factor f (between 0.35 and 1). Find the smallest f in steps of 0.1 that keeps the first-storey drift at no more than 1.5 times the largest drift of the other storeys. Justify the trade-off.

4 Session 4: The response-spectrum shortcut: base shear without a time history

A. Real-world problem

A design office has to check hundreds of buildings and cannot run a 2400-step time history for each one. The usual shortcut skips the time history: shake a family of simple one-degree-of-freedom oscillators with the ground record once, call the largest displacement of each one the *response spectrum*, then read off one number per mode and combine the modes. For the 10-storey frame of Session 3 (floors of 200 tonnes, storeys of 350 MN/m) the full run gives a peak base shear of 8.77 MN. How close does the shortcut get with three modes, and for which buildings does it go wrong? (Synthetic record and a made-up frame; a screening exercise, not a design method.)

B. Concept and theory

The response spectrum. Take an oscillator of period T and damping ratio 5 percent, shake it with the record a_g(t), and write down the largest displacement it ever reaches: S_d(T). Repeat for many periods. The pseudo-acceleration is S_a(T) = (2 pi / T)^2 S_d(T): a mass m on that oscillator feels at most the force m S_a at the peak. Dividing by the peak ground acceleration (2 m/s^2) says how many times the record is amplified at that period. A very stiff oscillator (T near 0) follows the ground, so S_a / PGA tends to 1; for this record the amplification peaks at 4.4 around T = 0.72 s and falls to 0.19 at 3 s.

One mode. Mode i alone would move the floors in the shape phi_i with peak amplitude Gamma_i S_d(T_i) (Gamma_i is the participation factor of Session 3). The forces that hold that shape are F_j = m_j Gamma_i phi_j,i w_i^2 S_d(T_i), and they add up to the base shear

V_i = M*_i S_a(T_i), with M*_i = Gamma_i^2 the effective modal mass.

The same force follows from the lowest storey, k_1 Gamma_i phi_1,i S_d(T_i), and the code checks that the two agree for all ten modes. A storey shear is the sum of the floor forces above the storey.

Combining modes. The modes do not peak at the same moment, so adding the V_i overestimates (the sum is an upper bound). The standard rule is the square root of the sum of squares (SRSS), V = sqrt(sum V_i^2), which assumes independent peaks and works when the periods are well separated. When two periods are close the modes peak together, and the complete quadratic combination (CQC) uses a correlation rho_ij between the modes: 1 for equal frequencies and close to 0 when they are far apart. With equal damping zeta and r = w_j / w_i, rho_ij = 8 zeta^2 (1 + r) r^1.5 / ((1 - r^2)^2 + 4 zeta^2 r (1 + r)^2). In a shear building the first two periods differ by a factor of 3, so CQC and SRSS agree here (8.983 against 8.979 MN).

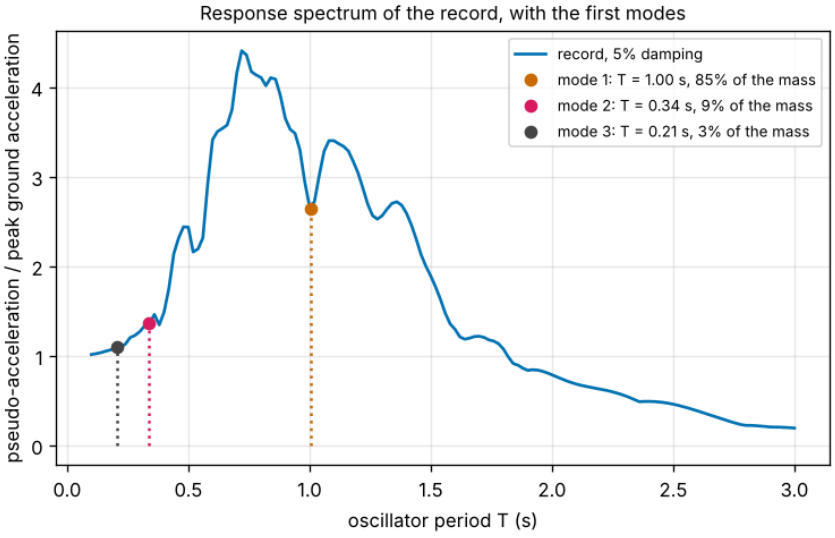
- T, w: period and angular frequency of a mode (s, rad/s)
- S_d(T): peak displacement of an oscillator of period T (5 percent damping) (m)
- S_a(T): pseudo-acceleration (2 pi / T)^2 S_d(T) (m/s^2)
- PGA: peak ground acceleration of the record (2.0) (m/s^2)
- Gamma_i, M*_i: participation factor and effective modal mass of mode i (kg^(1/2), kg)
- V_i: peak base shear of mode i, M*_i S_a(T_i) (N)
- V: combined base shear (SRSS, sum or CQC) (N)
- rho_ij: correlation of modes i and j in CQC (no unit)

$$\ddot{q} + 2\zeta\omega\dot{q} + \omega^2 q = -a_g(t), \qquad S_d(T) = \max_t |q|, \qquad S_a = \omega^2 S_d$$

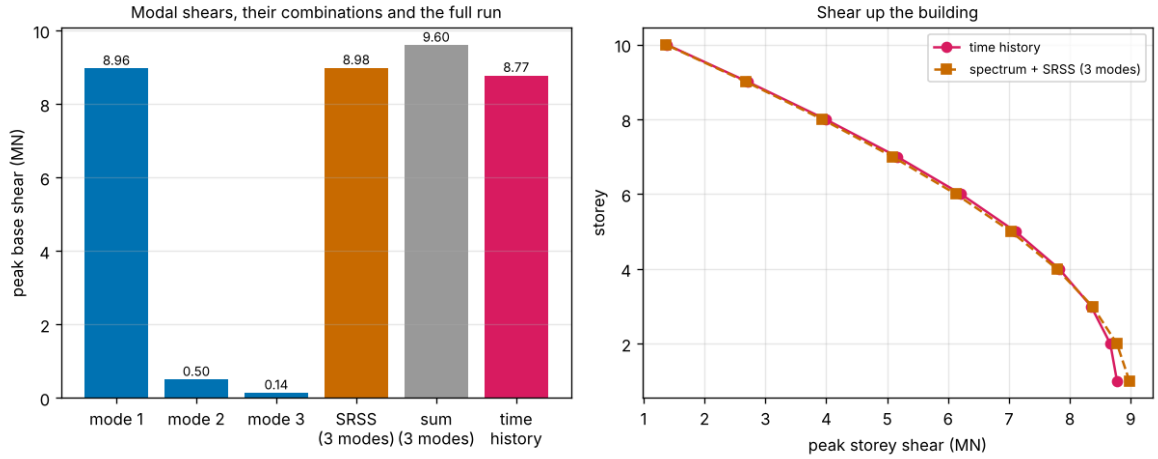
$$V_i = \omega_i^2 \, \Gamma_i \, (\boldsymbol{\phi}_i^{\mathsf{T}} M \mathbf{r}) \, S_d(T_i) = M_i^* S_a(T_i) = k_1 \Gamma_i \phi_{1,i} S_d(T_i)$$

$$V_{\mathrm{SRSS}} = \sqrt{\sum_i V_i^2}, \qquad V_{\mathrm{sum}} = \sum_i |V_i|, \qquad V_{\mathrm{CQC}} = \sqrt{\sum_i \sum_j \rho_{ij} V_i V_j}$$

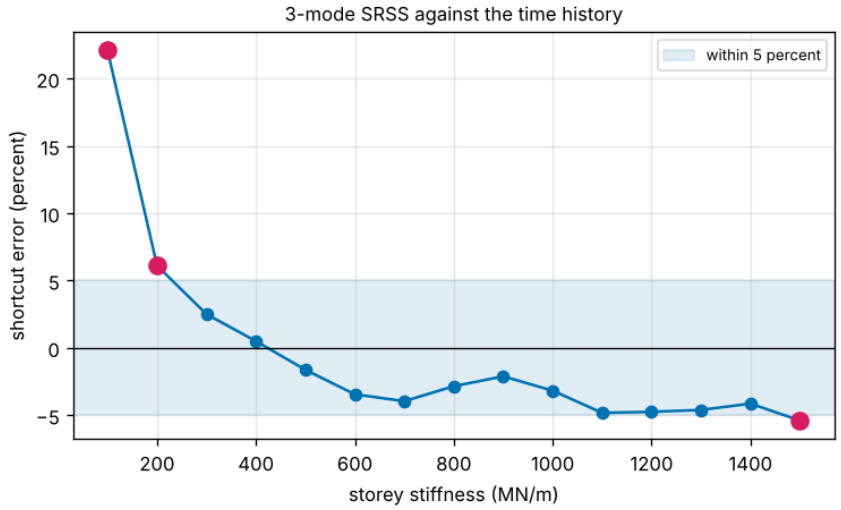
$$\rho_{ij} = \frac{8\zeta^2(1+r)r^{3/2}}{(1-r^2)^2 + 4\zeta^2 r(1+r)^2}, \quad r = \frac{\omega_j}{\omega_i}$$



The spectrum of the record with the first three modes marked; the first mode sits in a dip.



Left: modal base shears, SRSS, the plain sum and the time history. Right: peak storey shear up the building.



Error of the 3-mode SRSS shortcut against the storey stiffness; magenta points are outside 5 percent.

E. Worked example

Take the two-storey building of Session 3 (m = 200 000 kg, k = 3.5e8 N/m, T1 = 0.2430 s, T2 = 0.0928 s, effective mass shares 94.72 and 5.28 percent of the total mass 400 000 kg). A made-up spectrum gives S_a = 2.0 m/s^2 at T1 and 1.0 m/s^2 at T2. Find the modal base shears, the SRSS and the plain sum, and say how the numbers change for the 10-storey frame of Section A.

1. Effective masses: M*_1 = 0.9472 x 400 000 = 378 885 kg and M*_2 = 0.0528 x 400 000 = 21 115 kg.
2. Modal base shears V = M* S_a: V_1 = 378 885 x 2.0 = 757 771 N = 757.8 kN and V_2 = 21 115 x 1.0 = 21 115 N = 21.1 kN.
3. SRSS: sqrt(757.8^2 + 21.1^2) = sqrt(574 261 + 445) = 758.1 kN.
4. Plain sum: 757.8 + 21.1 = 778.9 kN, which is 2.7 percent above the SRSS because the second mode is small.
5. For the 10-storey frame the code reads S_a from the spectrum of the record: 5.29, 2.73 and 2.19 m/s^2 at T = 1.005, 0.337 and 0.206 s; with the effective masses 1 696 000, 183 000 and 61 800 kg this gives V = 8.964, 0.498 and 0.135 MN, an SRSS of 8.979 MN and a plain sum of 9.598 MN. The full time history gives 8.775 MN.

Answer. Two storeys: V = 757.8 and 21.1 kN, SRSS 758.1 kN, sum 778.9 kN. Ten storeys: SRSS 8.979 MN, which is 2.3 percent above the time-history peak of 8.775 MN, so the roadmap target of 5 percent is met; the plain sum is 9.4 percent above.

F. Questions

1. **Conceptual.** Why does the shortcut combine the modes with the square root of the sum of squares rather than adding the modal peaks, and when could SRSS be too small?
2. **Numerical.** A building of total mass 1.5e6 kg has a first mode that carries 80 percent of the mass and a second that carries 15 percent. The spectrum gives S_a = 3.0 m/s^2 at the first period and 4.0 m/s^2 at the second. Find the modal base shears, the SRSS and the plain sum.
3. **Graph and simulation interpretation.** In the spectrum figure the first mode (T = 1.00 s) sits at S_a / PGA = 2.64, in a dip, while the peak of 4.4 is at 0.72 s. A retrofit that stiffens every storey to 700 MN/m moves the first period to 0.71 s. What does the spectrum predict for the first-mode base shear, and what does the full time history give?
4. **Design and optimisation challenge.** The owner can choose a storey stiffness of 100, 200, ..., 1500 MN/m. The screening office accepts the shortcut only if it is within 5 percent. For which options can it use the shortcut, and what should it do for the others? Which failures are the dangerous ones?

5 Real-world mini-project

Is this 10-storey frame in danger from a 1 s ground period? Fix it by stiffening. A 10-storey frame has floors of 200 tonnes, storeys of 350 MN/m each, storey height 3.5 m and 5 percent damping. The city's dominant ground period is 1 s. First show that the frame is in danger: compute its first period and the amplification factor T1 / Tg from Session 2. Then use `peaks` from Session 3 to stiffen every storey by a factor from 1.0 to 4.0 in steps of 0.5. A stiffening passes when the peak roof sway is at most half of the original and the peak inter-storey drift ratio (drift divided by 3.5 m) is at most 0.5 percent.

When can the shortcut be trusted? A screening office wants to use the 3-mode SRSS shortcut instead of a time history for the 10-storey frame, and the owner is considering stiffer columns. For storey stiffnesses from 100 to 1500 MN/m in steps of 100 (floors of 200 tonnes, the record of Session 3), compute the first period, the time-history base shear and the 3-mode SRSS estimate. The shortcut passes for a stiffness if its error is within 5 percent. Use `error_vs_stiffness` and `passing` from Session 4, and say which failures are on the safe side (estimate above the time history) and which are not. **Deliverable:** A script that prints a table for the 15 stiffnesses (first period, time history, estimate, error, pass or fail), reports how many pass and the range of stiffnesses that pass, and saves a plot of the error against the stiffness with the 5 percent band. Add two sentences on what the screening office should do about the failures. **Success looks like:** 12 of the 15 stiffnesses pass, namely 300 to 1400 MN/m; 100 MN/m (T1 = 1.88 s, +22.1 percent) and 200 MN/m (+6.1 percent) fail on the safe side, and 1500 MN/m (-5.4 percent) fails on the unsafe side. Your write-up notes that the time-history base shear is not smaller for stiffer frames (15.3 MN at 700 MN/m against 8.8 MN at 350 MN/m), and that the pass range depends on this one jagged record, so it should not be read as a rule.

6 The path ahead

1. **Session 5: Rayleigh damping instead of modal damping.** A damping matrix built from two coefficients that give 5 percent at modes 1 and 3.

2. **Session 6: Tune the mass damper.** A search over damper tuning and damper damping, averaged over six random records.
3. **Session 7: A real record and real periods.** A loader for a public ground-motion record, with the dominant period found from its Fourier spectrum.
4. **Session 8: Yielding and torsion.** Elastic-perfectly-plastic storeys in the Newmark solver, and a plan-view model with an eccentric mass.

7 Answers and hints

Session 1

1. **Conceptual.** Hint: Think of a child on a swing: what changes if the child is heavier? Then think about when a push adds to the motion and when it works against it. Solution: A larger mass is harder to accelerate for the same spring force, so each swing takes longer; a stiffer spring pulls back harder for the same sway, so each swing is quicker. That is T = 2 pi sqrt(m / k). When the ground's period equals the building's, every shove arrives just as the building is already moving the same way, so the effects add up cycle after cycle and only damping limits the sway. Even small pushes can build up to a large sway: at 5 percent damping the gain at resonance is 10.
2. **Numerical.** Hint: w = sqrt(k / m), then T = 2 pi / w. Check the units: k / m is in N/m per kg = 1/s^2. Solution: w = sqrt(20 000 000 / 500 000) = sqrt(40) = 6.325 rad/s. T = 2 pi / 6.325 = 0.9935 s. That is within 1 percent of the 1 s ground period, so this building is almost at resonance and would be strongly amplified (about 10 times). Sanity check: it is heavier than the worked example (200 000 kg, T = 0.628 s), so its period is longer.
3. **Graph and simulation interpretation.** Hint: The ratio is r = T / Tg = building period over ground period. Use the formula or read the curve. Solution: With Tg = 2 s, r = 0.5 and the gain is 1.33: the building mostly follows the slow ground motion, a little amplified. With Tg = 0.5 s, r = 2 and the gain is 0.33: the shaking is too quick for the building to follow, so it hardly moves. Both are far from the resonance peak of 10 at Tg = 1 s.
4. **Design and optimisation challenge.** Hint: Invert T = 2 pi sqrt(m / k) to get k = m (2 pi / T)^2, then use r = T / Tg in the gain formula. Solution: For T = 0.5 s: k = 200 000 * (2 pi / 0.5)^2 = 3.158e7 N/m, r = 0.5, gain 1.33. For T = 1.5 s: k = 200 000 * (2 pi / 1.5)^2 = 3.509e6 N/m, r = 1.5, gain 0.79. Both avoid the resonance peak of 10. Downsides: the stiff design costs a lot more column material (9 times the stiffness of the flexible one) and attracts larger forces; the flexible design is cheaper but sways a long way in total (a long period means large displacements) and may be uncomfortable or hit the next building. Assumptions: the ground has one steady period and the model is one lump.

Session 2

1. **Conceptual.** Hint: Think of the three cases: pushing very slowly, at the natural rhythm, and very quickly. Solution: For r near 0 the push is so slow that the building just follows it statically (D = 1). Near r = 1 the push is in step with the building's own swing, so energy is added every cycle and only damping limits the growth, giving D about 1 / (2 zeta). For large r the push reverses before the heavy roof can respond, so the roof hardly moves (D < 1, and D falls as r grows). More damping removes energy faster and lowers the peak: 25, 10, 5 and 2.5 for damping ratios 0.02, 0.05, 0.1 and 0.2; it changes the curve little far from resonance.
2. **Numerical.** Hint: w = sqrt(k / m); r = T / Tg; u_max = (a0 / w^2) D. Solution: w = sqrt(4e6 / 1e5) = sqrt(40) = 6.325 rad/s; T = 2 pi / 6.325 = 0.9935 s; r = 0.9935; D = 1 / sqrt((1 - 0.987)^2 + (0.0994)^2) = 9.98; a0 / w^2 = 2 / 40 = 0.05 m; u_max = 0.05 * 9.98 = 0.499 m. Sanity check: r is almost 1, so D is almost 10 and the sway is almost half a

metre. Units: a0 / w^2 is (m/s^2) / (1/s^2) = m.

3. **Graph and simulation interpretation.** Hint: At r = 1 the formula collapses to 1 / (2 zeta). Solution: At r = 1: D = 25, 10, 5 and 2.5, which is 1 / (2 zeta) in each case. Halving the damping doubles the peak. At r = 1.5 and zeta = 0.05, D = 0.794: the building is shaken faster than its own rhythm and moves less than the static push would.
4. **Design and optimisation challenge.** Hint: Use r = 0.7107 and r = 1.2186 (from D = 2) with T = r Tg, then k = m (2 pi / T)^2 and u_max = (a0 / w^2) * 2. Solution: Stiff side: T = 0.7107 * 1.5 = 1.066 s, k = 6.948e6 N/m, sway = 0.115 m. Flexible side: T = 1.2186 * 1.5 = 1.828 s, k = 2.363e6 N/m, sway = 0.339 m. Both have D = 2, but the flexible building sways nearly three times further (0.339 against 0.115 m) because a0 / w^2 grows with the period, even though it needs much less stiffness. The stiff design keeps displacements small, which protects partitions and neighbours; it costs more column material and attracts larger forces. The flexible one needs a way to cope with large displacement. Assumptions: steady sine ground motion, one lump, 5 percent damping.

Session 3

1. **Conceptual.** Hint: Look at the mode shapes: in which mode do all floors move the same way? Then look at the effective masses. Solution: In mode 1 all floors move in the same direction, so the ground pushes all of them in step: the participation Gamma = phi^T M r is largest, and 84.8 percent of the mass (10 storeys) takes part. In mode 2 the top moves against the middle: pushes on the two groups of floors partly cancel and only 9.1 percent takes part. Mode 2 changes sign once along the height, so one floor sits near a node. Higher modes have more sign changes and less mass. That is why a handful of modes (here three, 97 percent) is enough.
2. **Numerical.** Hint: w^2 = lambda k / m with lambda = (3 -/+ sqrt 5) / 2, so k / m = 1000 per s^2. Solution: w1^2 = 0.382 * 1000 = 381.97 so w1 = 19.544 rad/s and T1 = 0.3215 s; w2^2 = 2.618 * 1000 = 2618.0 so w2 = 51.167 rad/s and T2 = 0.1228 s. Both periods are far shorter than 1 s, so the ratio T / Tg is below 0.33 for both and neither is near resonance; mode 1 (0.3215 s) is the closer one. Units: k / m is 1/s^2, w is in rad/s.
3. **Graph and simulation interpretation.** Hint: Compare the two bars in storey 1 and also look at the rest of the storeys. Solution: Storey 1: 95.3 mm with a soft first storey against 25.1 mm in the uniform frame, 3.8 times as much, as drift ratio 2.7 percent against 0.72 percent. The other storeys stay near their normal drift. The weak storey concentrates the bending: the whole building above it moves with it, so the first-storey columns take the damage. This is the typical failure of buildings with an open ground floor.
4. **Design and optimisation challenge.** Hint: Use `soft_storey_stiffness(factor=f)`, run Newmark and compare the first drift with the maximum of the others. Solution: At f = 0.6 the first-storey drift is 55.4 mm, 1.71 times the largest other drift: fails. At f = 0.7 it is 43.5 mm, 1.46 times: passes. So f = 0.7 (the first storey at least 70 percent as stiff as the others) is the smallest with margin; f = 0.35 gives 2.97 times. Trade-off: stiffening the ground floor costs structure and may attract more force; f = 1 gives 1.01 times, a uniform drift. Assumptions: this one synthetic record, linear behaviour, uniform masses.

Session 4

1. **Conceptual.** Hint: Do the modes reach their peaks at the same moment? What changes if two modes have almost the same period? Solution: Each mode oscillates at its own frequency, so their peaks occur at different times and partly cancel; adding the peaks assumes they all coincide and so gives an upper bound (+9.4 percent here for three modes, against +2.3 for SRSS). SRSS treats the peaks as independent, which is a good model when the periods are well apart, as in this frame (T1 / T2 = 3). If two periods are almost equal, the two

modes oscillate in step and peak together, so SRSS underestimates; CQC corrects this with a correlation rho_ij that tends to 1 for equal frequencies (the CQC of two equal-frequency modes is their plain sum).

2. **Numerical.** Hint: V = M* S_a with M* = share x total mass. Solution: M*_1 = 0.80 x 1.5e6 = 1.2e6 kg and M*_2 = 0.15 x 1.5e6 = 2.25e5 kg. V_1 = 1.2e6 x 3.0 = 3.6e6 N = 3.6 MN and V_2 = 2.25e5 x 4.0 = 0.9e6 N = 0.9 MN. SRSS = sqrt(3.6^2 + 0.9^2) = sqrt(13.77) = 3.71 MN. Sum = 4.5 MN, which is 21 percent above the SRSS. Units: kg times m/s^2 is newtons.
3. **Graph and simulation interpretation.** Hint: Compare S_a at the two periods, and remember that V = M* S_a with the same effective mass. Then compare 8.8 MN with the time-history base shear for 700 MN/m that the lesson prints in its stiffness table. Solution: At T = 0.71 s the spectrum is at 4.33 times the ground acceleration instead of 2.64, so S_a rises by a factor of 4.33 / 2.64 = 1.64. The effective mass share (84.8 percent) is the same for a uniform frame, but the first-mode base shear is 14.68 MN instead of 8.96 MN, and the SRSS is 14.69 MN. The full time history at 700 MN/m gives 15.29 MN, 1.74 times the 8.78 MN of the original frame (the error of the shortcut is -4.0 percent). Stiffening moved the frame into the strongest part of the spectrum of this record, so the force it must resist rose: stiffer does not mean less force.
4. **Design and optimisation challenge.** Hint: Use `error_vs_stiffness` and look at the sign of the error as well as its size: a negative error means the shortcut gives less than the time history. Solution: The shortcut is within 5 percent for 300, 400, ..., 1400 MN/m (12 of 15 options; errors from +2.5 to -4.8 percent). It fails for 100 MN/m (+22.1 percent), 200 MN/m (+6.1 percent) and 1500 MN/m (-5.4 percent). The first two overestimate, so a force limit set from the shortcut would be on the safe side, but they are too far off to be useful; the 1500 MN/m option underestimates, which is the unsafe direction, so it needs a time history. Even the passing options at 1100 to 1400 MN/m are close to the limit (-4.1 to -4.8 percent), so a careful office would run a time history for any option whose shortcut error is near the limit. One synthetic record decides these numbers; with a real design spectrum the pass range would be different.

8 How to join

Read `CONTRIBUTING.md` in the repository, then pick an issue labelled `good first issue`. Repository: https://github.com/KrunalPanchal1995/pan-iit-oss/tree/main/openbuildingsway